ORIGINAL ARTICLE

Adolescent Blood Pressure and Cardiovascular Disease Before Age 50 Years

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BACKGROUND: Adolescent blood pressure guidelines rely on expert consensus because evidence on cardiovascular outcomes is limited. This study aimed to examine the link between adolescent blood pressure indices and early cardiovascular events.

METHODS: We conducted a cohort study among 902 741 adolescents aged 16 to 19 years who were evaluated for mandatory service from 1979 to 2019, excluding those with preexisting cardiometabolic conditions. Individuals were followed until 50 or death or insurance loss or December 31, 2021, whichever occurred first. Exposures included baseline blood pressure and American Academy of Pediatrics categories: normal (<120/<80), elevated (120/<80–129/<80), stage 1 (130/80–139/89), stage 2 (≥140/90), and hypertension (clinical diagnosis). The primary outcome was incident cardiovascular events (ischemic heart disease or cerebrovascular disease). Hazard ratios were estimated using Cox models adjusted for demographic, socioeconomic, and clinical confounders.

RESULTS: During over 18 million person-years of follow-up, 6305 cardiovascular disease events were recorded, yielding an incidence rate of 0.35 per 1000 person-years. Increased diastolic, systolic, and mean arterial blood pressure were significantly associated with increased risk. Compared with the Normal group, adjusted hazard ratios for cardiovascular disease were 1.14 (95% CI, 1.08–1.22) for stage 1, 1.31 (1.20–1.44) for stage 2, and 2.42 (1.87–3.12) for hypertension. Risk in the stage 1 category was particularly sensitive to diastolic blood pressure.

CONCLUSIONS: Higher blood pressure indices during adolescence were strongly associated with an elevated risk of early cardiovascular disease, highlighting the potential need to refine current guidelines to better reflect cardiovascular risk.
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Key Words: adolescent ■ blood pressure ■ cardiovascular disease ■ hypertension ■ myocardial infarction

Cardiovascular disease (CVD) is a leading cause of morbidity and mortality worldwide.[1,2] Its rising prevalence among young adults (under age 50 years) raises public health concerns.[3] Addressing early risk factors can influence the disease's trajectory, and thereby reduce health care burden and improve health outcomes.[4,5]

Although elevated blood pressure (BP) is a major CVD risk factor in older adults (over age 50 years),[6] the link between adolescent BP indices and later CVD remains underinvestigated. Due to the limited data available to identify a specific level of BP that leads to CVD in young adulthood, current adolescent BP thresholds rely on expert opinion and extrapolation from adult studies.[7,8]

We used a unique data linkage between a cohort of almost 1 million adolescents who had recorded measurements of systolic BP (SBP) and diastolic BP (DBP), height, and weight, with a CVD registry from a state-mandated health provider (Maccabi Healthcare Services [MHS]). We aimed to investigate the relation between adolescent BP and the risk of CVD events before age 50 years.

METHODS

Design, Setting, and Population

This cohort study involved Israeli adolescents aged 16 to 19 years who underwent medical evaluations for mandatory

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NOVELTY AND RELEVANCE

What Is New?

This study links adolescent blood pressure to early cardiovascular events in 902 741 individuals who were systematically evaluated. It identifies a J-shaped risk relationship for diastolic blood pressure, finds a significant interaction between systolic and diastolic blood pressure, and provides crucial outcome data to inform thresholds.

What Is Relevant?

Stage 1 blood pressure values or higher (≥130/80 mm Hg) are an independent risk marker for ischemic heart disease and cerebrovascular disease, even after controlling for known confounders.

Clinical/Pathophysiological Implications?

Current adolescent blood pressure thresholds may benefit from a more outcome-driven approach, putting more emphasis on diastolic blood pressure.

Nonstandard Abbreviations and Acronyms

AAP	American Academy of Pediatrics
BMI	body mass index
BP	blood pressure
CVD	cardiovascular disease
DBP	diastolic blood pressure
HR	hazard ratio
i3C	International Childhood Cardiovascular Cohort
MAP	mean arterial pressure
MHS	Maccabi Healthcare Services
PP	pulse pressure
SBP	systolic blood pressure

military service between January 1, 1979, and December 31, 2019, and were subsequently insured by MHS. Exclusion criteria included a history of CVD; any cardiac, coagulation, kidney, or glycemic disorders diagnosed before or during the evaluation; and missing evaluation data on body mass index (BMI) or BP. Follow-up began at the baseline evaluation and ended on CVD diagnosis, the last day insured, death, age 50 years, or December 31, 2021, whichever occurred first. The Israel Defense Forces Medical Corps and MHS review boards approved this study. The article adheres to the STROBE reporting guidelines.[9] The data that support the findings of this study are available from the corresponding author on reasonable request.

Baseline Evaluation of BP

Physicians conducted baseline assessments, including interviews, medical history reviews, and physical examinations. Any concerning history or abnormal findings led to further tests or specialist consultations to establish a diagnosis.

The evaluation of BP in our cohort has been previously described.[10] Briefly, BP was measured after 5 to 10 minutes of rest in a seated position, using a cuff on the right arm at heart level. Mercury sphygmomanometers were used until 2007, when a digital device (Omron M2) became the primary method, measuring with a precision of 1 mm Hg. Two readings were taken, and their average was documented.[11]

DBP and systolic BP (SBP) were measured directly. Mean arterial pressure (MAP) and pulse pressure (PP) were derived from those measurements (MAP=1/3×SBP+2/3×DBP and PP= SBP-DBP). BP categories were based on the American Academy of Pediatrics (AAP) classification[7]: normal (<120/<80 mm Hg), elevated (120/<80–129/<80 mm Hg), stage 1 (130/80–139/89 mm Hg), stage 2 (≥140/90 mm Hg), and hypertension (from clinical diagnosis). Individuals initially in stage 2 were referred for 10 additional measurements over at least 3 weeks and ultimately included in stage 2 only if a clinical diagnosis of hypertension was excluded.

Certified physicians classified adolescents as having hypertension at baseline if the adolescents had a preexisting diagnosis of hypertension, were taking antihypertensive medications, or were diagnosed during the recruitment evaluation. This multistep process ensured that the hypertension category reflected persistent clinical hypertension rather than a transient phenomenon or measurement error.

Assessment of CVD

Israel Defense Forces conscription registry data were linked to the MHS registries. MHS covers 26% of the Israeli population, but about 40% of those are obligated to military service.[12] A comparison of baseline characteristics of adolescents evaluated for military service by later MHS membership was previously published.[12] Established in 1998, the MHS CVD registry auto-collects data from electronic medical records, hospitalization records, lab results, prescriptions, physiological signals (eg, electrocardiograms), and investigation reports.[13] Laboratory results are analyzed centrally.

CVD included ischemic heart disease and cerebrovascular disease. Ischemic heart disease comprised myocardial and nonmyocardial infarction (*International Classification of Diseases, Ninth Revision* codes: 410.x, 412, 429.7, 429.79, 36.x, 411.x, 414.x, 429.2). Cerebrovascular disease included ischemic, hemorrhagic, transient ischemic attack, and occlusion of cerebral arteries (*International Classification of Diseases, Ninth Revision* codes: 433.x1, 438.x, 435.x, 434.x). Procedure codes for interventions like percutaneous coronary intervention or coronary artery bypass grafting were considered as evidence of a CVD event.

Other Variables

Several variables were collected during the baseline assessment. Height and weight were measured with the adolescents barefoot and in their underwear.[14] BMI was calculated as weight in kilograms divided by height squared in meters. Residential socioeconomic status was classified as low (1–4), medium (5–7), and high (8–10)[15,16] (Supplemental Methods). Education levels reported by the Ministry of Education were categorized as complete (11 or more years) or partial. Country of birth was classified as either Israel or abroad, according to the Ministry of the Interior. Cognitive scores, associated with cardio-metabolic outcomes,[17–19] were grouped as low (1–3), medium (4–7), and high (8–9; Supplemental Methods). Unimpaired health at baseline was defined as the absence of any medical condition that requires chronically prescribed medications or medical follow-up (eg, hypothyroidism with chronic levothyroxine use), a history of cancer, or major surgery.[20–22]

Data on incident diabetes, hypertension, chronic kidney disease, and cancer were sourced from dedicated MHS registries described previously.[12] Smoking status, derived from a dedicated registry, was categorized as ever-smoker or never-smoker. Ever-smoker was defined as having any indication of smoking in their medical record (Supplemental Methods).

Statistical analysis

Cox proportional hazards models with age as the timescale were used to estimate CVD hazard ratios (HRs) across BP values and groups. Interactions were assessed with an interaction term and an F-test. The proportional hazards assumption was visually confirmed. Fully adjusted Cox models were adjusted for sex, BMI, socioeconomic status, cognitive score, educational level, country, year of birth, and smoking status.

For each continuous BP measure (DBP, SBP, MAP, and PP), we fitted a penalized spline Cox model with 4 df to visualize the hazard function and its 95% CIs.[23] Individuals with clinically diagnosed hypertension were excluded from this specific analysis because their values may have been altered by antihypertensive therapy. An additional model using a tensor-product smooth examined the interaction between DBP and SBP in their association with CVD.

To probe residual confounding and ascertainment bias, we repeated the spline analyses with incident cancer as a negative-control outcome.[24] We also applied the same models to 2 prespecified secondary outcomes: ischemic heart disease and cerebrovascular disease, to facilitate comparison with studies of more specific cardiovascular end points. For continuous exposures that showed a linear association with CVD, *Z* scores were calculated, and the association was estimated using a 1-standard-deviation measure.

The Normal BP group was the reference for the AAP BP categories. The primary categorical analyses involved the crude association, the fully adjusted model, and a partially adjusted model omitting BMI from the adjustment set.

The robustness of findings was assessed through additional analyses: (1) restriction to individuals with unimpaired health at baseline; (2) restriction to individuals who were not diagnosed with diabetes, hypertension, or chronic kidney disease by the end of follow-up; (3) stratification by smoking status; (4) limitation of the age of onset to 40 or 35 years; (5) stratification by sex; (6) assessment of secondary outcomes, including ischemic heart disease and cerebrovascular disease; (7) restriction to individuals evaluated since 1995, when near-complete CVD capture became possible due to the establishment of the registry; and (8) consideration of death as a competing risk (Fine-Gray model).[25]

Data were also analyzed using the BP categories used in the Young Swedish Men cohort to enable comparisons of incidence and HR.[26]

A complete case analysis was performed. All the tests were 2-tailed. Correlations were calculated using Pearson (r_p) and Spearman (r_s) correlation coefficients. The data were analyzed using R version 4.4.2 and the packages tidyverse, survival, mgcv, ggsurvfit, gtsummary, and forester.

RESULTS

Overall, 956 828 adolescents were evaluated during the study period. Of these, 54 087 (5.7%) were excluded from the analysis (Figure 1). The final cohort comprised 902 741 adolescents (497 631 male, 55.1%).

Normal, elevated, stage 1, stage 2, and hypertension comprised 394 355 (43.7%), 147 069 (16.3%), 309 555 (34.3%), 49 594 (5.5%), and 2168 (0.2%) adolescents, respectively. The mean age at study entry was 17.3 (0.5) years and was similar across the BP groups (Table). The mean BMI and the prevalence of high cognitive scores increased from the normal to the hypertension category. The prevalence of adolescents born in Israel was lower in the hypertension group. Baseline characteristics by sex were also provided (Table S1).

Sex did not modify the association between BP and the risk of CVD (*P* interaction of 0.37, 0.27, and 0.07, with DBP, SBP, and BP group, respectively). During 18 064 299 person-years of follow-up, with a median follow-up of 19.8 years (interquartile range, 11.3–29.9 years), 6305 CVD events were recorded. The reported number of incident CVD events from normal to hypertension were 1760 (0.4%), 907 (0.6%), 2835 (0.9%), 740 (1.5%), and 63 (2.9%), respectively. The corresponding rates were 2.3, 3.3, 4.3, 6.7, and 14.5 per 10 000 person-years (Table S2).

Kaplan-Meier survival analysis demonstrated a graded pattern of higher CVD risk, progressing from individuals with normal BP to those with clinically diagnosed hypertension, especially after age 40 years (Figure 2).

A spline model with DBP exhibited a J-shaped association with CVD (linear $P<0.001$, nonlinear $P<0.001$), with a strong linear component above 60 mm Hg. Spline models with SBP or MAP revealed a positive linear association with CVD (SBP linear $P<0.001$, nonlinear $P=0.34$; MAP linear $P<0.001$, nonlinear $P=0.56$). There was no evidence of an association between PP and CVD (linear and nonlinear *P* values of 0.42 and 0.30, respectively; Figure 3). Restricted cubic spline models yielded near-identical results.

DBP and SBP showed a moderate correlation (rp=0.47, rs=0.47), and their interaction model partially

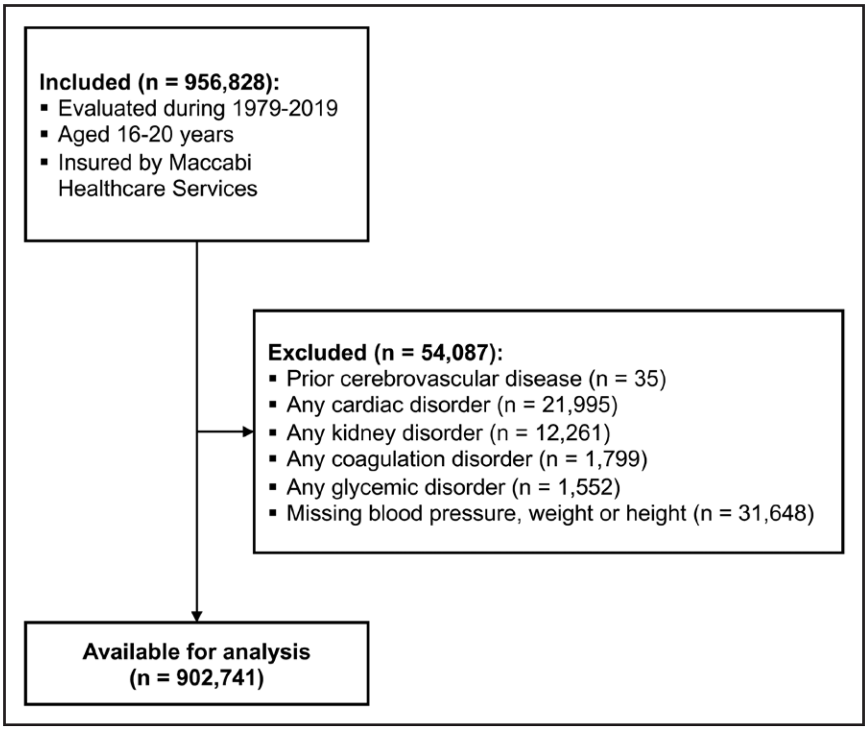

Figure 1. Flowchart of cohort inclusion and exclusion criteria. Some adolescents met multiple exclusion criteria. Thus, the total number of individuals excluded is smaller than the simple summation of measures.

aligned with AAP BP categories (Figure 4; *P* interaction <0.001). In the stage 1 category, HRs ranged from 1.0 to 1.5 across DBP values. There was no evidence of an association between BP and incident cancer (considered as a negative control outcome; Figure S1).

Spline models also demonstrated associations between BP measurements and secondary outcomes. DBP, SBP, and MAP were associated with ischemic heart disease, and DBP was associated with cerebrovascular disease (Figures S2 and S3).

In a crude Cox regression, with *Z* scores of BP measurements showing a linear association, a 1-SD increase in DBP, SBP, and MAP was associated with HRs of 1.21 (1.18–1.24), 1.29 (1.26–1.32), and 1.31 (1.28–1.34), respectively. The respective fully adjusted HRs were 1.09 (1.06–1.12), 1.08 (1.05–1.11), and 1.09 (1.07–1.13).

Compared with the normal group, the crude HRs for CVD in the elevated, stage 1, stage 2, and hypertension groups were 1.33 (95% CI, 1.22–1.44), 1.53 (1.44–1.62), 2.12 (1.94–2.31), and 5.48 (4.27–7.05), respectively. When adjusting for all the factors except BMI, the HRs were 1.10 (1.02–1.19), 1.25 (1.18–1.33), 1.58 (1.44–1.72), and 3.76 (2.92–4.84). The fully adjusted HRs were 1.04 (0.96–1.13), 1.14 (1.08–1.22), 1.31 (1.20–1.44), and 2.42 (1.87–3.12; Figure 5).

The above results persisted in 4 additional analyses. The first was of adolescents with unimpaired health at baseline (Figure S4). The second analysis was of adolescents who were not diagnosed with various incident diabetes, hypertension, or chronic kidney disease during the interval between the baseline examination and the end of follow-up (Figure S5). The third analysis was performed after stratifying the cohort by smoking status (Figure S6; *P* interaction=0.1). The fourth analysis considered the outcome as CVD diagnosed before age 35 years (Figure S7). Compared with women in the normal group, women in the stage 2 and hypertension groups had significantly higher HRs of 1.43 (1.08–1.89) and 6.77 (2.77–16.50), respectively (Figure S8A). Subgroup analyses by BMI, country of birth, educational level, and residential socioeconomic status are also available (Figures S8B through S8E).

Regarding the secondary outcomes, compared with the normal group, stage 1, stage 2, and hypertension were associated with significantly higher HRs for ischemic heart disease, of 1.20 (1.12–1.28), 1.37 (1.24–1.51), and 2.19 (1.64–2.92), respectively. Hypertension alone was associated with a higher adjusted HR for cerebrovascular disease of 3.25 (1.89–5.59; Figure S9).

When the cohort was limited to those evaluated since 1995, point estimates were similar but with wider CIs (Figure S10). Similar results were observed in a competing risk model (Figure S11). Comparing the men in this study with the Young Swedish Men cohort, using the same BP groups, yielded similar cumulative incidence (Figure S12; Table S3) and HR (Table S4). Comparing the adjusted and unadjusted HRs of 1 SBP *Z* score between this study and a similar i3C (International Childhood Cardiovascular Cohort) study also yielded similar results (Table S5).

DISCUSSION

In this nationwide cohort of Israeli adolescents followed for almost 2 decades, we found an association between

Table. Cohort Baseline Characteristics (n=902 741) by Blood Pressure Groups

Characteristic	Normal (<120/<80 mm Hg), n=394 355	Elevated (120/<80–129/<80 mm Hg), n=147 069	Stage 1 (130/80–139/89 mm Hg), n=309 555	Stage 2 (≥140/90 mm Hg), n=49 594	Hypertension (clinically diagnosed), n=2168
Age, examination	17.3 (0.4)	17.3 (0.5)	17.3 (0.5)	17.4 (0.5)	17.6 (0.6)
Sex, men	162 672 (41.3%)	93 216 (63.4%)	201 441 (65.1%)	38 325 (77.3%)	1977 (91.2%)
Body mass index, kg/m²	21.1 (3.0)	22.0 (3.5)	22.2 (3.7)	23.2 (4.3)	26.8 (5.3)
Residential socioeconomic status					
Low	74 184 (18.9%)	29 674 (20.3%)	62 392 (20.3%)	11 117 (22.6%)	460 (21.3%)
Medium	200 895 (51.3%)	74 441 (51.0%)	159 619 (52.0%)	24 659 (50.1%)	1144 (53.0%)
High	116 427 (29.7%)	41 860 (28.7%)	84 976 (27.7%)	13 396 (27.2%)	554 (25.7%)
Unknown	2849	1094	2568	422	10
Cognitive score					
Low	48 132 (12.3%)	19 448 (13.3%)	39 735 (12.9%)	6483 (13.2%)	279 (13.0%)
Medium	283 731 (72.5%)	103 015 (70.6%)	217 356 (70.8%)	34 687 (70.6%)	1465 (68.0%)
High	59 461 (15.2%)	23 480 (16.1%)	49 772 (16.2%)	7946 (16.2%)	409 (19.0%)
Unknown	3031	1126	2692	478	15
Complete education	369 172 (93.9%)	136 424 (93.0%)	284 222 (92.1%)	45 244 (91.5%)	1998 (92.4%)
Unknown	1165	363	892	149	6
Born in Israel	327 755 (83.1%)	121 980 (83.0%)	254 637 (82.3%)	41 618 (83.9%)	1581 (73.0%)
Unknown	101	35	90	12	1
Smoking, ever	134 196 (34.0%)	48 194 (32.8%)	103 985 (33.6%)	15 725 (31.7%)	673 (31.0%)

Categorical variables are presented as n (%) and continuous variables as mean (SD).

higher BP levels at 16 to 19 years of age and incident CVD before age 50 years. These observations were consistent across DBP, SBP, and MAP. Compared with AAP-defined normal values (<120/<80 mm Hg), individuals with Stage 1 or higher values (≥130/≥80 mm Hg) had a significantly higher risk, even after excluding key

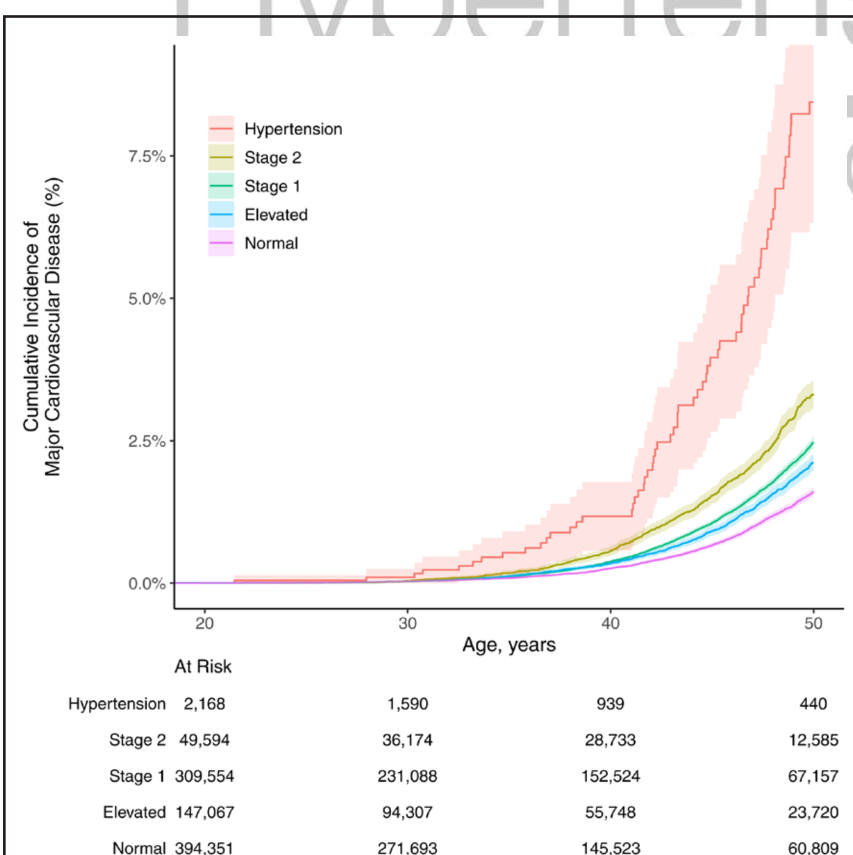

Figure 2. The cumulative incidence of cardiovascular disease by adolescent blood pressure group.
This figure shows the cumulative incidence of cardiovascular disease by age and adolescent blood pressure group. The *x* axis represents age in years, and the *y* axis indicates the cumulative incidence of cardiovascular disease as a percentage. The number of individuals at risk in each blood pressure group is shown at 4 ages (20, 30, 40, and 50 years). The numbers declined over time as individuals were censored or developed cardiovascular disease. Using a log-rank test, the global *P* value was <0.001. The Bonferroni-adjusted pairwise *P* value between all groups was <0.001, except for stage 1 compared with elevated, for which it was *P*=0.002.

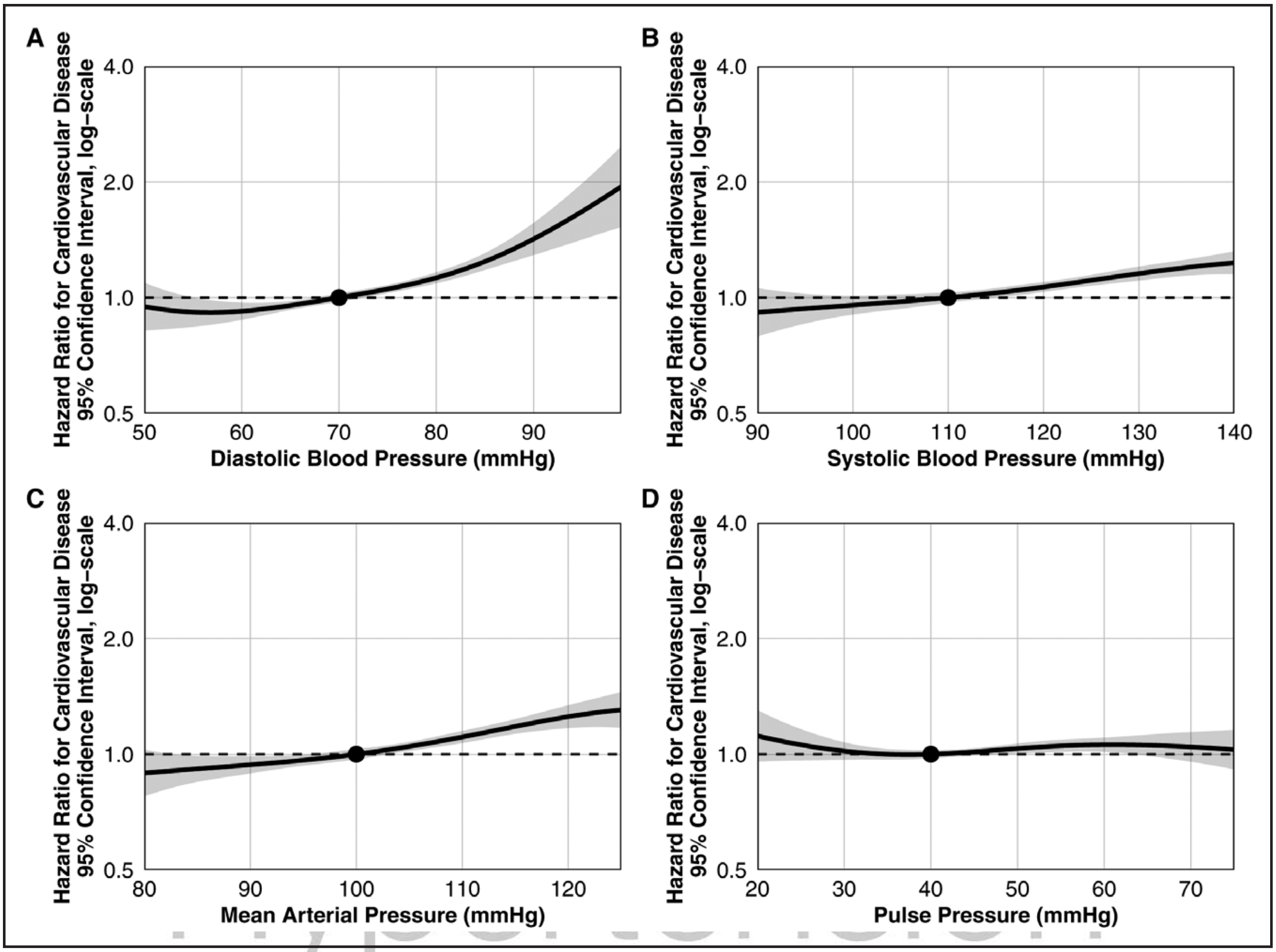

Figure 3. Spline models of the association between various blood pressure measures during adolescence and cardiovascular disease risk before age 50 years.

This figure illustrates the adjusted hazard ratios for cardiovascular disease across diastolic blood pressure (**A**), systolic blood pressure (**B**), mean arterial pressure (**C**), and pulse pressure (**D**). The adjusted hazard ratios are presented on a logarithmic scale, with 95% CIs indicated by the shaded regions. The reference values are 70, 110, 100, and 40 mm Hg for diastolic blood pressure, systolic blood pressure, mean arterial pressure, and pulse pressure, respectively. **A** through **C**, Demonstrate that elevated blood pressure during adolescence is associated with a higher risk of cardiovascular disease in adulthood.

comorbidities and adjusting for potential confounding factors, including BMI and smoking status. For adolescents with clinically diagnosed hypertension compared with their normotensive counterparts, the risk of CVD was over 2-fold.

Current AAP adolescent BP cutoffs rely primarily on expert consensus and are extrapolated from adult data because outcome-based evaluations of continuous BP indices are limited.[7,8] A Swedish study examined a boys-only cohort and reported higher CVD risk across adolescent BP categories, including those with elevated levels that do not meet the diagnostic threshold criteria for hypertension.[26] Another study linked childhood SBP *Z* scores and CVD in middle age.[27] Using continuous analysis of BP values and their joint distribution, our study demonstrated that current AAP-defined categories capture much of the risk associated with higher BP but could be further refined by placing greater emphasis on DBP, particularly in the stage 1 category, where HRs ranged from 1.0 to 1.5 across DBP values.

Studies that link adolescent BP and incident CVD among adolescent girls are rare.[27] In our study, we found no significant interaction by sex. However, a subgroup analysis among 405 110 girls revealed that only those with stage 2 or clinically diagnosed hypertension were at higher risk, possibly due to the lower power to investigate girls only. Importantly, adolescent girls with clinically diagnosed hypertension had nearly a 7-fold higher risk of developing CVD than did girls with normal BP, highlighting a potential population for a high-yield intervention.

Our findings are likely generalizable to other populations for several reasons. First, findings on BP and hypertension observed in independent adolescent cohorts from different countries were replicated in our cohort, yielding similar results. These include the link to end-stage renal disease[28,29] and the link to cardiovascular mortality.[30,31]

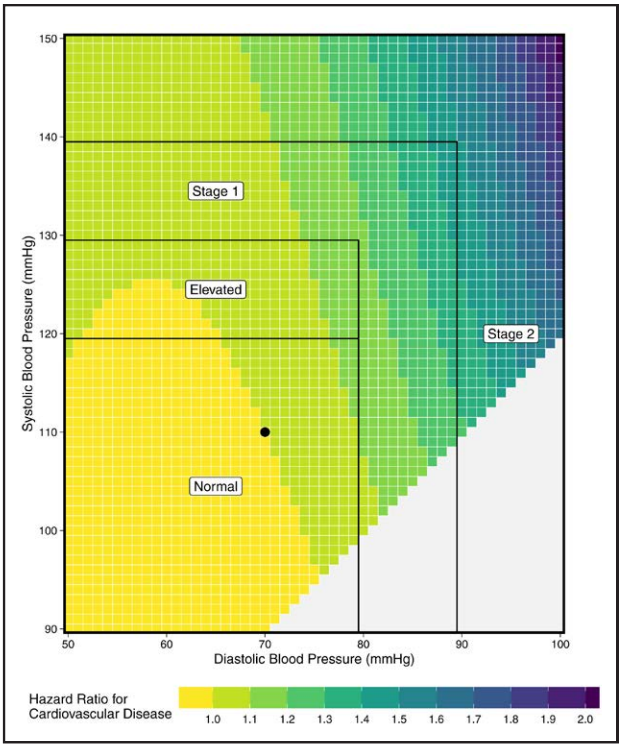

Figure 4. Spline model of the interaction between diastolic and systolic blood pressure in association with cardiovascular disease risk before age 50 years.
This heatmap illustrates the adjusted hazard ratios for cardiovascular disease across the interaction between diastolic and systolic blood pressure in adolescents, and demonstrates misalignment between the American Academy of Pediatrics blood pressure categories and the hazard function interpolated from the outcome data. The black dot is the reference blood pressure value (110/70 mm Hg).

Second, in the present study, we directly compared our findings to those from the Swedish adolescent cohort[26] and the i3C study.[27] Despite differences in population structure and health care systems, we observed strikingly similar cumulative incidences and HRs. Third, the Israeli population is genetically and ancestrally diverse, reflecting broad geographic origins, including Europe, North Africa, the Middle East, and the former Soviet Union.[32] This heterogeneity supports the applicability of our findings beyond a single ethnic or national context.

Elevated BP since early life likely leads to cumulative vascular damage, including endothelial dysfunction, accelerated atherosclerosis, carotid intima-media thickening, increased left ventricular afterload, and cardiac remodeling.[33] These conditions are exacerbated by the persistence of high BP into adulthood.[34] The resolution of elevated BP has been shown to reverse at least some of the damage.[35] The J-shaped association between adolescent DBP and incident CVD found in our study, with a strong linear component above 60 mm Hg, mirrors a similar finding in a cohort of Koreans aged 20 years-39 years during a routine health examination.[36] This suggests that lower DBP may compromise coronary and cerebral perfusion even in youth. The linear association between MAP and incident CVD, as well as the consistent null findings for PP, support the growing recognition of MAP as a physiologically integrative hemodynamic marker in young individuals. Furthermore, the results may support current hypotheses that endothelial injury, rather than arterial stiffening (and thus wide PP), is the primary driver of early CVD.

This study has several limitations. First, BP was evaluated on a single occasion during adolescence, implying that the true underlying associations between adolescent BP and CVD events in early adulthood are even stronger than those reported here.[37] Despite this, our results suggest that BP readings on even a single occasion in adolescence remain associated with CVD later in life. Second, we lacked serial data on potential confounding factors, suggesting the possibility of residual confounding in our results. Third, data on physical exercise and diet were lacking. However, this omission may have been partially mitigated by adjusting for BMI, which is highly affected by both. Fourth, we lacked data on medication use, including in those with clinically diagnosed hypertension. Fifth, we lacked data on whether hypertension was primary or secondary. Although our registry does not distinguish between primary and secondary hypertension, we excluded adolescents with possible secondary causes, such as preexisting kidney, cardiac, glycemic, and coagulation disorders, and adjusted for other risk factors, such as obesity, to isolate as much as possible the association of increased BP during adolescence and CVD in early adulthood. Lastly, there was potential for misclassification in individuals who died from CVD before diagnosis, which could have diluted the strength of our results. This type of misclassification was unlikely, as most young adult deaths result from injuries, and unexplained deaths undergo thorough investigations. Moreover, consistent results from the competing risks analysis suggested that, although misclassification is possible, its effect is likely small. More generally, the validity of our findings is suggested by their consistency across sensitivity analyses; the similarity of CVD incidences and HRs we observed for men to those reported in prior studies; and our negative control outcome analysis.

Conclusions

This study linked higher BP values among adolescent boys and girls to CVD events before age 50 years. The risk was significantly evident even after controlling for key CVD risk factors. The results suggest that current cutoffs are associated with risk but could be refined to reflect CVD risk better.

Perspectives

The findings from the extensive cohort study indicate that high BP in late adolescence is an independent

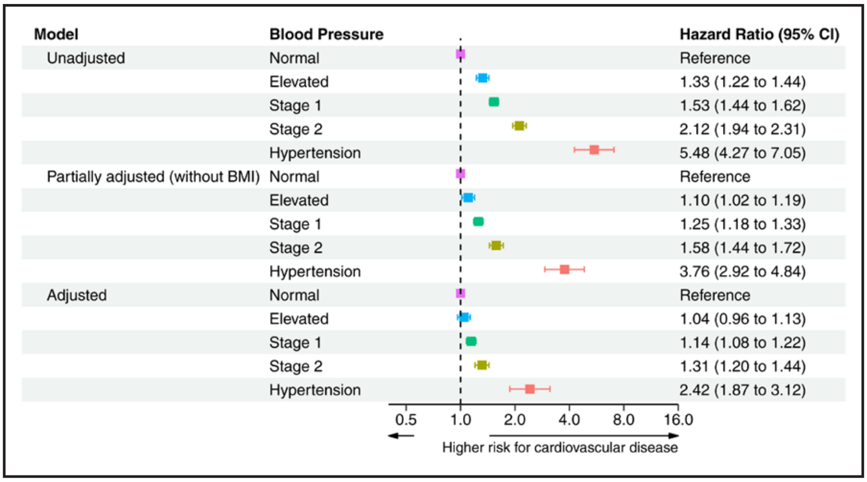

Figure 5. Adjusted and unadjusted hazard ratios for cardiovascular disease by blood pressure group.
This figure displays unadjusted, partially adjusted, and adjusted hazard ratios for the association between adolescent blood pressure categories and the risk of cardiovascular disease before age 50 years. The reference line at a hazard ratio of 1.0 indicates the risk for individuals in the Normal blood pressure group (<120/<80 mm Hg). The *x* axis represents hazard ratios on a logarithmic scale, with higher values indicating a higher risk of cardiovascular disease. The data reveal a graded increase in cardiovascular risk from the normal group to stage 2 (≥140/90 mm Hg), with even higher hazard ratios for individuals clinically diagnosed with hypertension. BMI indicates body mass index.

risk marker for cardiovascular events before age 50. Even Stage 1 BP levels (130/80–139/89 mm Hg) are linked with a significant rise in HRs for ischemic heart and cerebrovascular disease. Additionally, the J-shaped relationship between adolescent diastolic BP and cardiovascular events, along with the interaction between DBP and SBP and these events, suggests that current reliance on expert consensus and adult studies for adolescent BP thresholds may benefit from a shift toward a more refined outcome-driven approach. As global rates of early-onset CVD continue to increase, these findings highlight the importance of adolescence as a crucial period for cardiovascular risk assessment.

ARTICLE INFORMATION

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